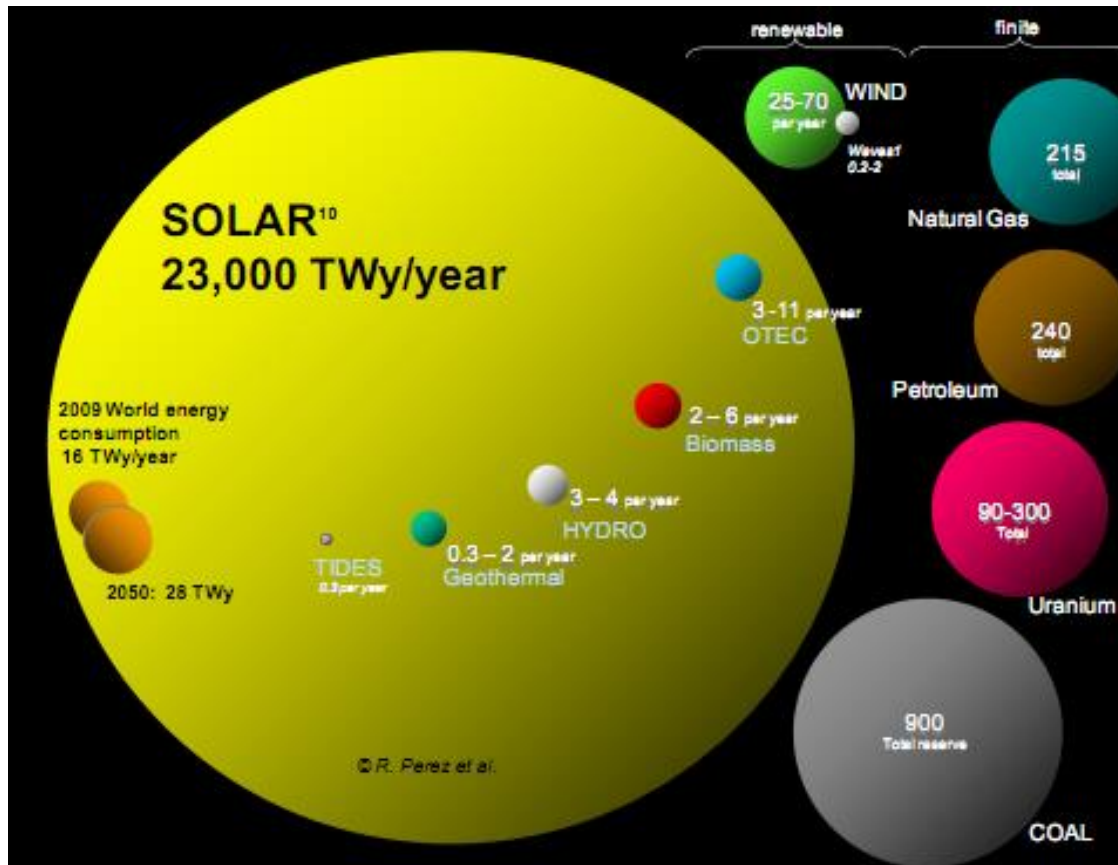


4 MORE ENERGY THAN WE'LL EVER NEED

Once you discover how much energy we get from the sun, thinking in terms of energy scarcity becomes kind of absurd. I love this visualization that I've adopted from Perez et al. ([Perez, 2022](#)).



The big one is the yearly energy from the sun. Wind is also abundant compared to our energy needs but all other renewable energy flows are relatively modest. At first glance it might seem as if the coal ball is pretty big too. But these are three dimensional balls remember. If the yearly energy from the sun was a basketball, coal would be a tennis ball. And that basketball is one year of sun while that tennis ball is all coal.

If the sun was a basketball, human energy use would be comparable to a walnut. Just picture it in your mind lying on the table next to each other. Basketball and walnut. It's literally nuts how much energy we get from the sun!

In numbers: the sun delivers 173,000 terawatts of power continuously to Earth ([Perez, 2022](#)) while humanity uses around 18 terawatts in 2024 ([Energy Institute, 2025](#)). That is a ratio of roughly 10,000 to one.

Finally, and then I'll stop: in one hour, the sun delivers more energy to Earth than humanity consumes in a year (Lewis, 2006). Best-in-class panels can now convert more than a quarter of sunlight directly into electricity (Liu, 2024). Lab records for tandem cells have exceeded 34%, and the theoretical limit for multi-junction designs reaches above 50% (Sun, 2025). The efficiency of solar panels is still increasing. It's *by far* the energy source with the most potential here on earth.

Wind is also abundant. The sun heats the atmosphere unevenly, creating pressure differentials that drive air movement. The resulting wind energy is a tiny fraction of the energy of solar radiation, but it carries roughly 100 times more energy than humanity needs. Even if we just harvested 20% available at good sites, it could still give us all we need (Archer, 2005).

Scientists have known this for over a century. They were not secret. But for most of that time, they were academic curiosities, because we had no practical way to capture diffuse sunlight at scale. The learning curves described in the previous chapters changed that.

The problem, it turns out, was never scarcity. The problem was that we were capturing energy in an extraordinarily roundabout way.

4.1 The Long Way Around the Sun

Consider what we have been doing for the past two centuries.

Hundreds of millions of years ago, plants captured sunlight through photosynthesis and stored it in their tissues. When those plants died, some were buried under sediment before they could decompose. Geological pressure transformed that ancient biomass, over tens of millions of years, into coal, oil, and natural gas.

Today we locate these deposits, often kilometres underground or beneath the ocean floor. We drill or dig. We extract the material, sometimes expending enormous energy just to bring it to the surface. We transport it across oceans in ships that themselves burn fossil fuels. We refine it. Finally, we burn it.

The burning releases heat. We use that heat to boil water. The steam spins a turbine. The turbine generates electricity. Along the way, we lose 60 to 70 percent of the energy as waste heat (EPA, 2025).

And all along, the sun kept shining. Every day. Every hour. Ten thousand times more energy than we needed, arriving without extraction, transportation, or combustion.

This was not stupidity. Burning was simply the only game in town. It's only recently that we've learned to capture sunlight directly and efficiently.

4.2 Photons to Electrons

Solar panels still seems like magic to me. One hundred and fifty million kilometres away, across an unimaginable vast nothingness, this giant fireball in the sky is fusing hydrogen

atoms into helium, and sending out bright light in the process. The sun hurls that light, which according to Einstein consists of tiny particles called photons, towards us over that unimaginably large distance.

This shower of photons powers all life on earth. If it hits a solar panel the result is a process that is astoundingly simple to understand and use. Basically the solar panel turns the energy in the photons into electricity just by lying in the sun.

In a bit more detail: the photons strike a semiconductor, typically silicon. Each photon is like a small packet of energy. When that packet of energy hits an electron in the silicon, it can knock that electron loose. The loose electron flows through a circuit. That flow is electricity ([Al-Ezzi, 2022](#)).

Sunlight in, electricity out. No combustion, no steam, no turbines. No intermediate steps involving ancient plants, geological transformation, extraction, or refinement. No transport all over the earth. Just photons hitting silicon anywhere on earth where there's sunshine.

Once the panel is manufactured and installed, the fuel is free. It will be free tomorrow. It will be free for the panel's thirty-year lifespan ([Jordan, 2016](#)). No one can embargo sunlight. No cartel can restrict supply. No war can threaten the shipping lanes.

There is no waste stream. A coal plant produces ash, sulfur dioxide, nitrogen oxides, particulate matter, mercury, and carbon dioxide ([EIA, 2024](#)). A solar panel produces nothing. It just sits there converting photons to electrons, decade after decade.

There is almost no maintenance. No moving parts in the panel itself. Panels installed in the 1970s still generate electricity today, albeit with some degradation ([Jordan, 2013](#)).

This simplicity is why solar rides such a steep learning curve. When something uses little material because it needs no fuel and can be produced and installed one small unit at a time, improvement compounds faster. The learning of solar is closer to the learning we know from computing than the learning we know from nuclear and coal.

I've got a roof consisting entirely of solar panels and they deliver twice the energy I need year round (including the electricity for my heat pump and my electric vehicle). In the gloomy Netherlands! Put a little panel on a roof in Africa, attach a small battery and an LED light, and a child can do homework at night, without a kerosene lamp that is poisoning her lungs and that her parents constantly have to buy expensive fuel for. It is worth pausing on how extraordinary this is.

Wind turbines work differently, capturing kinetic energy from moving air rather than electromagnetic energy from photons. But they share the essential feature: once built, the fuel is free, and the wind blows everywhere without anybody able to block it.

4.3 From Stock to Flow

Fossil fuels are a stock with no meaningful inflow: grandmother's savings, draining fast. But sunlight is different. It's not a stock. It's a flow: 173,000 terawatts arriving continuously,

whether we harvest it or not. The faucet is always open. It has been open for five billion years and will stay open for five billion more.

This fundamentally changes our relationship with energy. For most of human history, energy scarcity was not a mindset. It was an accurate description of reality. The inflow and stock were both very limited. There was only so much wood in the forest, only so much grain from each harvest. If your neighbor took more, you had less. Zero-sum thinking emerged because the world was, in fact, mostly zero-sum.

When we discovered fossil fuels, we made ourselves believe that we had found a hack out of that scarcity. The stored sunlight of millions of years allowed humanity to escape the immediate constraints of the land. But we drain down our bathtub full of fossil fuels at an alarming rate while the inflow is negligible. And some countries have lots of it while others have very little. Deep down we always knew that we were spending savings, not earning income. That the bathtub would run dry. That we were living on borrowed time.

Solar and wind are flows that we cannot deplete. If your neighbor installs solar panels, it does not reduce the sunlight falling on your roof. Energy is no longer a competition over fixed resources. Both of you can harvest the sun.

That doesn't mean it's a done deal that this will create human flourishing in harmony with nature. I wouldn't have written this book if I thought that. Building the infrastructure to capture solar energy requires materials, labor, and capital. Status quo interests will make the transition difficult and contested. New stakeholders will try to control us in new ways. Conflicts will not disappear simply because energy becomes abundant.

But the fundamental constraint has shifted. The resource itself, the energy, exists in quantities that dwarf any conceivable human demand. The sun does not need to be discovered, drilled, or transported. It is already there.

The teacher who showed us the burning candle in 1974 was not wrong about the danger. Fossil fuel depletion was real. Climate change was real. What he did not know was that another option was gestating in laboratories, too expensive to matter at the time, but riding a learning curve that would make it cheaper than coal within fifty years.

The scarcity that seemed inevitable has become optional. That is not an opinion. It's simply what the arithmetic shows and in the rest of this book I will explain it in detail.

We thought we were running out. Instead, we were just capturing energy in a very roundabout way. But are these solar panels for real? Most status quo experts still doubt that. So I made a visualisation to show how wrong they are.